



SpaceX next-gen Mars rocket

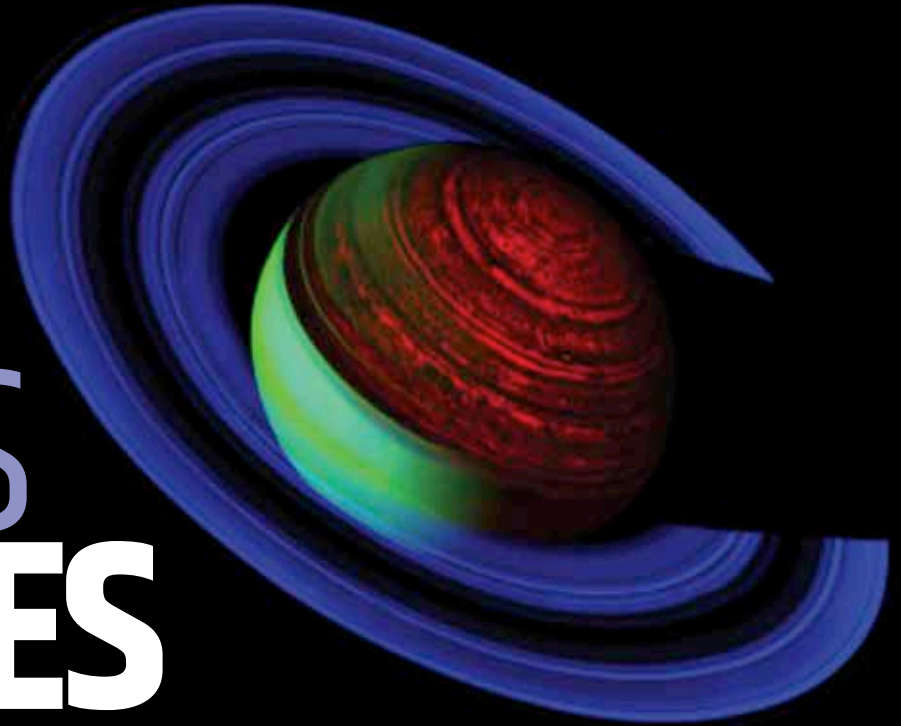
Eric Garcetti, mayor of LA, announced that SpaceX will build its new "BFR" spaceship at the new facility in the port of LA. <https://dgit.in/spacexmars>



NASA's Spacecraft: 3D-printed parts

NASA's Orion is being manufactured with 3D-printed parts which will be ready to embark on flights within a few years. <https://dgit.in/nasaorion>

WHY NASA EDITS IMAGES



The breathtaking photos of space released by NASA are often heavily post processed, but there are valid reasons why this is done

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ASA releases a steady stream of awe-inspiring images from its various astronomical instruments.

However, most of the photos are manipulated, and are very different from the RAW images captured by the telescopes. For this article, we will be focussing on four of the telescopes and instruments by NASA, the Hubble Space Telescope, the Curiosity Rover, the Cassini mission to Saturn, and the Juno mission to Jupiter. The underlying principles though, on how the images are captured, and why they are manipulated in a particular way, apply to most of the images released by NASA.

Almost every image released by NASA involves multiple exposures.

The images are captured in black and white, to get the most detail and information in the particular wavelengths. These images are then layered on to create a composite colour image. It is necessary to adjust the objects in the image if it has moved between the multiple exposures. At other times, it is necessary to create a mosaic from multiple shots. At times, these mosaics are digitally mapped on to a sphere to create an image of the planet or moon. Flyovers of terrains involve creating depth maps from the mosaics of planetary surfaces. After all this, there is colour correction done, portions of the image are touched up, and finally we get the images that we see in the newspapers and on the internet.

The astronomical instruments can view the objects they are observing in ultraviolet light, infrared light, and the visible light. Humans cannot see beyond the visible portion of the

spectrum, so the false colour images are adjusted accordingly, with the ultraviolet and infrared features pushed into the visible range.

The cover image for this article is actually a composite of 25 different images, taken over a period of thirteen hours, by the Cassini spacecraft. The human eye can see between 0.4 microns to 0.7 microns wavelengths. In this image the wavelengths of 2.3, 3.0 and 5.1 microns are represented by the visible blue, green and red images. All the data used in the image is from the infrared portion of the spectrum. The green portion is the day side, while the red portion is the night side. The night side is lit up by the thermal radiation produced deep within the atmosphere of the gas giant. At 2.3 microns the planet appears dark while the rings are bright, while at 3.0 microns, the rings are dark and the planet appears bright. This is because of the way the ice particles in the rings and methane in the Saturnian atmosphere absorb or reflect light at various wavelengths. In the composite image, the data from 2.3 microns is used for